Data Storytelling

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Fall 2026

Week 03

Announcements

- Homework 1 will be assigned next week. Will be due one week after that (week 05).

Data Analysis I

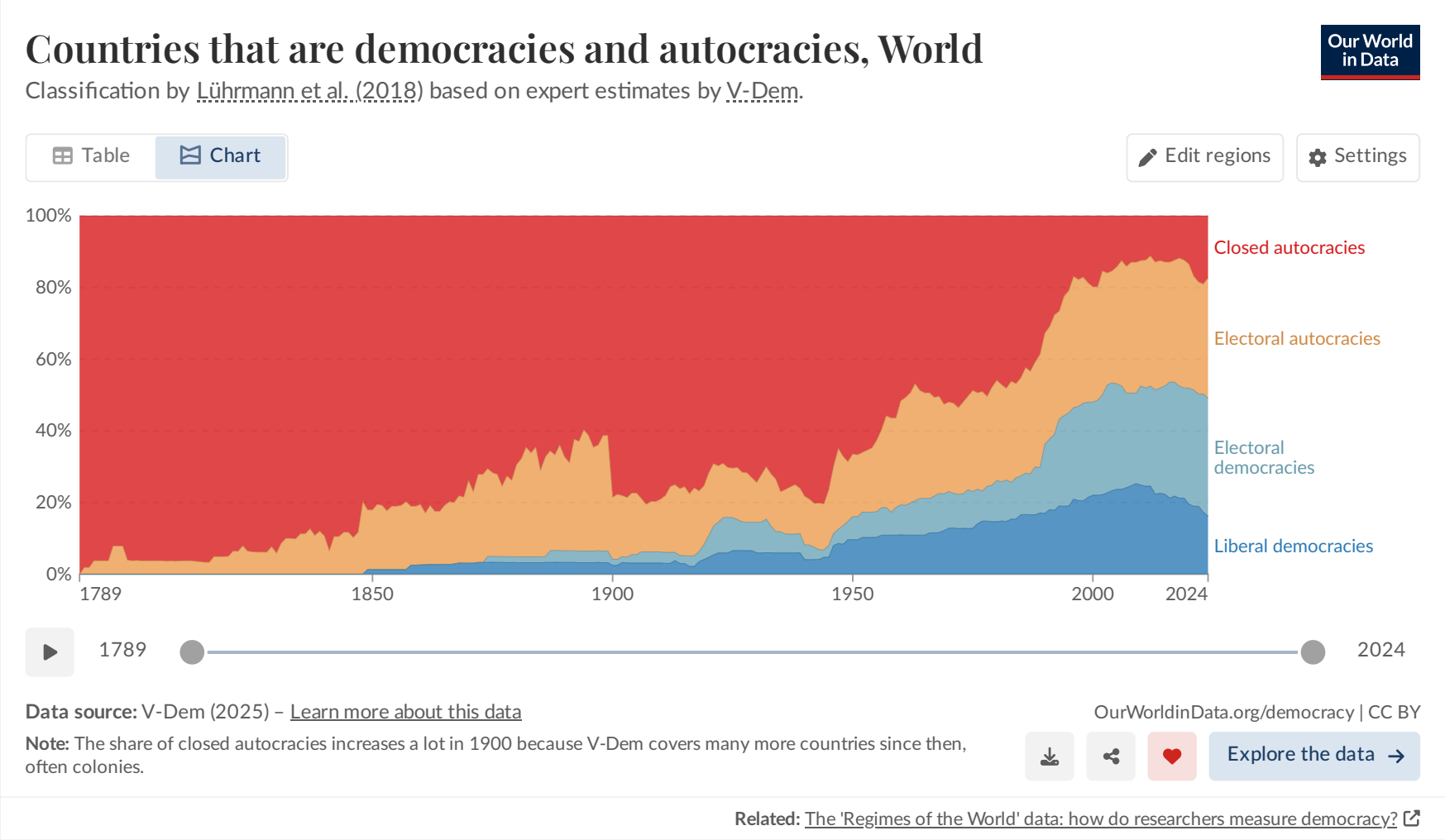
Side note

- Why start with data analysis? Why not start with data collection or data visualization?
- Data analysis is the process of making sense of data and extracting insights from it.
- Without a clear understanding of what your data is, what it represents, and a clear objective for your analysis, it will be difficult to craft any meaningful visualizations or to know what data you need to collect in the first place.

What is the objective of your analysis?

- **Measurement**:
  - How can we quantify a concept or phenomenon?
  - *Methods*:
    - Surveys (ex. World Values Survey, Afrobarometer)
    - Expert coding (ex. Polity IV, V-Dem)
    - Text analysis (ex. word counts, word embeddings, topic modeling)
  - *Examples*:
    - How has the number of democracies/autocracies changed in the world since 1789?

Measurement Example: V-Dem's Electoral Democracy Index



What is the objective of your analysis?

- **Prediction**:
  - Can we use existing data to predict events outside of the training data?
  - *Methods*:
    - Supervised Machine learning (ex. random forests, neural networks)
    - Statistical modeling (ex. regression)
  - Examples:
    - Can we predict the likelihood of a country experiencing a coup based on economic indicators?
    - Can we predict the outcome of an election based on polling data?

Prediction Example: 2024 Election Forecast

What is the objective of your analysis?

- **Explanation**:
  - How can we determine whether there exists a relationship between two or more variables?
  - *Methods*:
    - Statistical modeling (ex. regression, correlation)
    - Causal inference (ex. matching, instrumental variables, difference-in-differences)
  - Examples:
    - Does economic development lead to democratization?

Explanation Example: GDPPC and Democratization

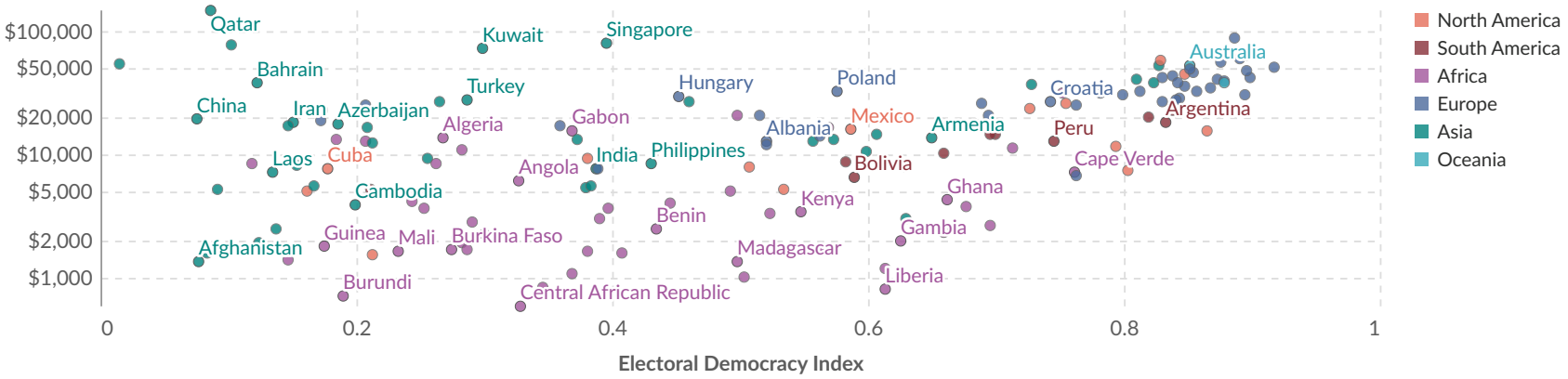
GDP per capita vs. Electoral Democracy Index, 2022

Our World in Data

GDP per capita is adjusted for inflation and differences in living costs between countries. The electoral democracy index is by V-Dem, based on expert estimates, and ranges from 0 to 1 (most democratic).

Table | Chart | Select countries and regions | Settings

GDP per capita (constant 2011 international-$; plotted on a logarithmic axis)



1789 — 2022

Data source: Bolt and van Zanden – Maddison Project Database 2023; V-Dem (2025) – Learn more about this data

Note: GDP per capita is expressed in international-$ at 2011 prices.

OurWorldinData.org/democracy | CC BY

Explore the data →

Related: How do different approaches score democracy across the world?

How do you do analysis?

- A long time ago, data analysis was done with pen and paper.
  - There was an entire industry around performing calculations, such as regression, by hand.
  - “Computers” were people!
- Not long ago, we started using software like SPSS, Stata, and Excel.
  - Faster, more consistent than earlier methods, but still limited in flexibility and reproducibility.
- Today, we use open source programming languages like R and Python to analyze data.

Importance of Coding

- Why use code to analyze data?
  - *Reproducibility:* Code allows others to reproduce your analysis and verify your results.
  - *Efficiency:* Code can automate repetitive tasks and handle large datasets more efficiently than manual methods.
  - *Flexibility:* Code allows for more complex analyses and custom visualizations that may not be possible with point-and-click software.

Why R?

- Has large open source community with many packages for data manipulation, analysis, and visualization.
- Quarto and R Markdown allow you to create dynamic documents that combine code, text, and visualizations in a single file.
  - Fun fact, all slides for this course were created using Quarto!
- R is free and open source, making it accessible to everyone.
  - Back in my day.. We learned Stata and SPSS in undergrad. I had to pay for those licenses out of pocket.
- As a result, it is widely used by statisticians and in the social sciences, including political science, sociology, economics, and psychology.

Let's get started with R!

- First, I'll go over a quick tutorial on how to clone our class GitHub repository
- Then, we'll go through some basic R syntax and explore the different types of objects in R
- Finally, we'll go over a tutorial on how to wrangle data in R using the `dplyr` package, which is part of the `tidyverse` collection of packages for data manipulation and visualization.
- We'll save Laudet and Imai's Estimating Causal Inference with Randomized Experiments for next week